Problem 2

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Problem 1 *For each of the following functions find an equation of the tangent line at the given point.*

(a) $f(x) = -5x^2 + 7x - 9 \quad at\ x = 3.$

Explanation. The slope of tangent line is given by this limit: $f'(3) = \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3}$. Notice that this limit has form $\frac{0}{0}$, so it is indeterminate.

$$\begin{aligned} f'(3) = \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3} &= \lim_{x \to 3} \frac{(-5x^2 + 7x - 9) - (-45 + 21 - 9)}{x - 3} \\ &= \lim_{x \to 3} \frac{-5x^2 + 7x - 9 + 33}{x - 3} \\ &= \lim_{x \to 3} \frac{-5x^2 + 7x + 24}{x - 3} \\ &= \lim_{x \to 3} \frac{(x - 3)(-5x - 8)}{x - 3} \\ &= \lim_{x \to 3} (-5x - 8) \\ &= -5(3) - 8 = -23. \end{aligned}$$

Point on tangent line: $(3, f(3)) = (3, -33)$.

An equation of tangent line is:

$$y + 33 = -23(x - 3)$$

(b) $g(u) = \sqrt{5u - 4} \quad at\ u = 3.$

Explanation. The slope of tangent line is given by: $g'(3) = \lim_{h \to 0} \frac{g(3 + h) - g(3)}{h}$.

Notice that this limit has form $\frac{0}{0}$, so it is indeterminate.

$$
\begin{aligned}
g'(3) = \lim_{h\to 0} \frac{g(3+h)-g(3)}{h} &= \lim_{h\to 0} \frac{\sqrt{5(3+h)-4}-\sqrt{11}}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{5(3+h)-4}-\sqrt{11}}{h} \cdot \frac{\sqrt{5(3+h)-4}+\sqrt{11}}{\sqrt{5(3+h)-4}+\sqrt{11}} \\
&= \lim_{h\to 0} \frac{5(3+h)-4-11}{h\left(\sqrt{5(3+h)-4}+\sqrt{11}\right)} \\
&= \lim_{h\to 0} \frac{5h+15-15}{h\left(\sqrt{5(3+h)-4}+\sqrt{11}\right)} \\
&= \lim_{h\to 0} \frac{5}{\left(\sqrt{5(3+h)-4}+\sqrt{11}\right)} \\
&= \frac{5}{\sqrt{5(3+0)-4}+\sqrt{11}} \\
&= \frac{5}{2\sqrt{11}}.
\end{aligned}
$$

Point on tangent line: $(3, g(3)) = (3, \sqrt{11})$.

Equation of tangent line:

$$y - \sqrt{11} = \frac{5}{2\sqrt{11}}\,(u-3)$$

(c) $s(z) = \dfrac{z}{z-5}$ *at* $z = 3$.

Explanation. The slope of tangent line is given by: $s'(3) = \lim_{z\to 3} \dfrac{s(z)-s(3)}{z-3}$.

Notice that this limit has form $\dfrac{0}{0}$, so it is indeterminate.

$$
\begin{aligned}
s'(3) &= \lim_{z\to 3} \frac{s(z)-s(3)}{z-3} \\
&= \lim_{z\to 3} \frac{\frac{z}{z-5} - \frac{3}{3-5}}{z-3} \\
&= \lim_{z\to 3} \frac{\frac{z}{z-5} + \frac{3}{2}}{z-3} \\
&= \lim_{z\to 3} \frac{\frac{2z}{2(z-5)} + \frac{3(z-5)}{2(z-5)}}{z-3} \\
&= \lim_{z\to 3} \frac{\frac{2z+3z-15}{2(z-5)}}{z-3} \\
&= \lim_{z\to 3} \frac{5z-15}{2(z-5)} \cdot \frac{1}{z-3} \\
&= \lim_{z\to 3} \frac{5(z-3)}{2(z-5)} \cdot \frac{1}{z-3} \\
&= \lim_{z\to 3} \frac{5}{2(z-5)} \\
&= \frac{5}{2(3-5)} = -\frac{5}{4}.
\end{aligned}
$$

Point on tangent line: $(3, s(3)) = (3, -3/2)$.

Equation of tangent line:

$$y + \frac{3}{2} = -\frac{5}{4}(z-3)$$